05-21 第2篇

阅读理解（说明文） How Does a Vacuum Flask Keep Water Hot for Hours?

① On a cold winter morning, many Chinese families fill a tall bottle with boiling water and put it on the kitchen table. Six or eight hours later, when a hungry child comes home from school, the water inside is still hot enough to make instant noodles. This clever bottle is called a **vacuum flask**, or simply "thermos". Why can a small bottle do what a big pot cannot?

② To answer the question, we first need to know how heat leaves hot water. Scientists say heat moves in three ways: by **conduction** (through solid things), by **convection** (through moving air or water), and by **radiation** (by invisible waves, like the warmth we feel from a fire without touching it). A normal cup does a very poor job of stopping these three ways, so the water inside gets cold quickly. A vacuum flask, however, is designed to block all three at the same time.

③ Inside a vacuum flask there are actually **two thin glass bottles**, one inside the other. Between the two glass walls, almost all the air has been pumped out. This empty space is called a **vacuum**. Heat has great difficulty crossing a vacuum, because without air there is nothing for the heat to travel through. In this way, both conduction and convection are almost stopped.

④ But heat can also leave by radiation, like tiny invisible light. To block it, the inner surfaces of the two glass walls are covered with a shiny silver layer. This silver coating acts like a small mirror. When the heat tries to escape as waves, the shiny surface reflects it back into the water, again and again. As a result, very little heat can get out.

⑤ The top of the bottle is closed by a thick cork or rubber stopper. Cork and rubber are poor heat carriers, so they also help to hold the warmth inside. Finally, the whole flask is protected by a colourful metal or plastic cover, which makes it safer to carry and harder to break.

⑥ A good vacuum flask is simple but clever. Two glass walls, a vacuum, a shiny coating and a soft stopper together turn a small bottle into a silent warm friend. Next time you hold a cup of hot tea made from flask water, remember: science is not only in laboratories — it is also in the kitchen.

根据短文内容，选择最佳答案。

1. What is the main topic of the passage?

A. How to make Chinese tea.
B. How a vacuum flask keeps water hot.
C. How to buy a cheap bottle.
D. Why water boils at 100°C.

2. According to paragraph ②, which is NOT one of the three ways heat leaves hot water?

A. Conduction.
B. Convection.
C. Radiation.
D. Reflection.

3. Why does the vacuum between the two glass walls help keep the water hot?

A. Because the vacuum stores extra heat.
B. Because without air there is almost nothing for the heat to travel through.
C. Because the vacuum heats the water again.
D. Because the vacuum changes the water into ice.

4. What is the job of the shiny silver layer inside the flask?

A. To make the flask look beautiful from outside.
B. To reflect heat waves back into the water so little heat can escape.
C. To change the colour of the water.
D. To stop light from the sun.

5. What does the writer want us to understand at the end of the passage?

A. Science only happens in big laboratories.
B. Science also lives in everyday things like a flask in the kitchen.
C. Flasks are not useful at all.
D. We should stop drinking hot water.

答案与解析

1.
【答案】B
【题型】主旨大意题
【定位】全文围绕 "How a vacuum flask keeps water hot"
【解析】文章解释保温瓶保温原理。A/C/D 均不是主题。故选 B。

2.
【答案】D
【题型】细节理解题

【定位】第②段 "conduction... convection... radiation"
【解析】三种方式分别为 A/B/C。D "Reflection"（反射）未被列为热的传递方式。故选 D。

3.
【答案】B
【题型】细节理解题
【定位】第③段 "without air there is nothing for the heat to travel through."
【解析】真空中无介质传热。A/C/D 均与科学原理相反。故选 B。

4.
【答案】B
【题型】细节理解题
【定位】第④段 "the shiny surface reflects it back into the water, again and again."
【解析】银层像镜子把热辐射反射回水中。A/C/D 均无据。故选 B。

5.
【答案】B
【题型】推理判断题 / 主旨大意题
【定位】第⑥段 "science is not only in laboratories — it is also in the kitchen."
【解析】科学存在于日常生活中。A 与原文相反；C/D 均无据。故选 B。

词汇注释（超纲词）

- **vacuum flask** [n.] 真空保温瓶
- **conduction** [n.] 传导
- **convection** [n.] 对流
- **radiation** [n.] 辐射
- **stopper** [n.] 瓶塞